Smart City / Mobility Project Background (Diriyah)

I. Project Background

Diriyah is a national-level cultural flagship project of Saudi Arabia and one of the core transformation initiatives under Saudi Vision 2030. Its objective is to create a world-class cultural tourism and lifestyle destination. It is not merely a real estate or infrastructure development project, but a national vehicle for cultural and economic structural transformation, and a key symbol of Saudi Arabia's future urban form and global cultural influence.

In terms of scale and development stage, Diriyah's total planned investment exceeds SAR 236Bn, with a development area of approximately 14 square kilometers. By 2030, the target annual visitor volume is 50M+, and as of 2025 visitor numbers have already reached 3.6M+, indicating a clear growth trajectory. The overall master plan includes 40 luxury hotels, 400+ retail brands, 1.6M+ square meters of office space, and multiple cultural landmarks including an Opera House, Arena, Museum, and Grand Mosque. This means its structure already contains a complete set of "city functional modules," rather than being merely a tourism zone or commercial complex.

From a development perspective, Diriyah is transitioning from the "Master Plan planning and construction phase" into the "scaled operational phase." It has completed core infrastructure and physical space construction, established asset-level operational optimization systems, secured international brand partnerships and global exposure, and built ESG and sustainability frameworks. These milestones indicate that its physical city construction and brand architecture are largely in place.

The key inflection point is that Diriyah has shifted from being a "construction project" to entering a "live city operations phase," yet its digital capabilities remain at the level of asset management and localized system optimization, without forming a city-scale intelligent operating structure.

II. Client Current Status

1 Existing Digital Capability Structure

At present, Diriyah's digital capabilities are built upon several foundational systems, but these systems remain function-centric rather than city-integrated.

First, at the foundational operations level, IBM provides the Smart Operations Backbone, primarily based on the IBM Maximo Application Suite asset management system, enabling real-time monitoring and predictive maintenance. This system focuses on asset and facility operations optimization to ensure efficient and stable infrastructure performance. However, its limitation is clear: it essentially optimizes physical assets only and does not provide city-level intelligent decision-making capabilities, nor does it constitute a Smart City strategic layer. It is an asset management platform, not a city intelligence platform.

Second, in the mobility domain, Diriyah has introduced Kakao Mobility for a Smart Parking pilot, enabling intelligent parking and localized mobility optimization. This represents a single-scenario mobility optimization effort. However, it also has structural limitations: it is a point solution pilot rather than an integrated city-scale mobility intelligence system, and it does not extend into visitor experience or cross-asset integration.

2 Current Structural Gaps

Structurally, Diriyah already possesses relatively mature infrastructure and energy capabilities, high-standard brand and asset structures, and a relatively complete operations and maintenance management system. However, its gaps are equally evident.

First, it has not yet established a digital twin or city operating system layer. Second, it lacks cross-asset data integration and a unified intelligence layer, resulting in data remaining siloed across multiple systems. In addition, it has not developed predictive city management capabilities and therefore cannot achieve forward-looking orchestration and resource optimization.

At the experience level, there is no immersive cultural interaction interface, nor an automation or future AI city capability framework. Fundamentally, the issue is that Diriyah is currently a "high-quality physical city + fragmented digital systems," rather than a "computable, predictable, orchestratable intelligent city."

III. Core Client Needs

Diriyah is not seeking a single system procurement, but rather the construction of a "City-Scale Intelligence Layer." In other words, they require a city-level AI Intelligence Platform that serves as Diriyah's unified intelligent core.

This requirement can be broken down into four layers.

1 City-Level Digital World Model

The first layer of need is to build a computable city digital twin system. This system must integrate asset, traffic, energy, visitor, event, and commercial data to achieve unified cross-system modeling.

Target capabilities include real-time city state visualization, city behavior simulation, visitor flow prediction, major event stress testing, and infrastructure load simulation. The essence of this layer is to make the city a “modelable object.”

2 Predictive Urban Intelligence

The second layer is to upgrade the operational model from reactive management to predictive orchestration. That is, not merely responding to issues, but forecasting them in advance and proactively orchestrating resources.

This includes visitor flow prediction, traffic pressure forecasting, event capacity simulation, energy demand forecasting, safety risk early warning, and dynamic optimization of ESG and sustainability metrics.

The goal is to elevate Diriyah from an “operated city” to a “predictable city.”

3 Immersive Cultural and Visitor Experience Interface

Diriyah is fundamentally a “cultural city,” not simply an infrastructure project. Therefore, the third layer of need is to build an immersive cultural interaction interface.

This includes digitization of cultural narratives, immersive experience layers, AR/XR city interaction, personalized visitor path recommendations, intelligent guide systems, and a digital cultural asset framework.

The goal is to create a true “Immersive Cultural Interface,” enabling the city itself to become an interactive cultural medium.

4 Future Automation Readiness

The final layer of need is to prepare for automation toward 2030 and beyond.

This includes city-level AI orchestration, autonomous scheduling capabilities, intelligent resource allocation, a unified data middle platform, and an AI agent collaboration system. It must also support intelligent operations capabilities for the 2030 Expo.

The ultimate objective is to become the first AI-native city operating model in the Middle East